WORKED SOLUTION

STEP I 1999, Question 7

Pure • Second-Order Differential Equations

STEP I 1999, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q7

The key idea

The function $\sin(k\sin^{-1}x)$ looks awkward to differentiate twice, but it becomes tame once we write $x = \sin\theta$, so that $y = \sin k\theta$. Multiplying the first derivative by $\sqrt{1-x^2}$ before differentiating again keeps the algebra short. For the last part, the point is that a second-order differential equation with a given value and gradient at one point has only one solution, so the polynomial we find *must* be $\sin(3\sin^{-1}x)$.

Solution

The differential equation

Let $y = \sin(k\sin^{-1}x)$ for $-1 < x < 1$. By the chain rule, using $\dfrac{\mathrm{d}}{\mathrm{d}x}\sin^{-1}x = \dfrac{1}{\sqrt{1-x^2}}$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k\cos(k\sin^{-1}x)}{\sqrt{1-x^2}}, \qquad \text{so} \qquad \sqrt{1-x^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = k\cos(k\sin^{-1}x).$$

Differentiating this second form with respect to x (product rule on the left, chain rule on the right):

$$\sqrt{1-x^2}\,\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{x}{\sqrt{1-x^2}}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{k^2\sin(k\sin^{-1}x)}{\sqrt{1-x^2}} = -\frac{k^2y}{\sqrt{1-x^2}}.$$

Multiplying through by $\sqrt{1-x^2}$, which is non-zero for $-1 < x < 1$, gives

$$(1-x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - x\frac{\mathrm{d}y}{\mathrm{d}x} + k^2y = 0,$$

which is $(*)$, as required.

The cubic solution when $k = 3$

Put $y = Ax^3 + Bx^2 + Cx + D$. The condition $y = 0$ at $x = 0$ gives $D = 0$, and $\frac{\mathrm{d}y}{\mathrm{d}x} = 3$ at $x = 0$ gives $C = 3$. Now

$$y' = 3Ax^2 + 2Bx + C, \qquad y'' = 6Ax + 2B,$$

and substituting into $(*)$ with $k^2 = 9$:

$$(1 - x^2)(6Ax + 2B) - x(3Ax^2 + 2Bx + C) + 9(Ax^3 + Bx^2 + Cx + D) = 0.$$

This must hold for all x, so we compare coefficients:

$$\begin{aligned} x^3: &\quad -6A - 3A + 9A = 0 \quad \text{(true for every } A\text{)},\\ x^2: &\quad -2B - 2B + 9B = 5B = 0,\\ x^1: &\quad 6A - C + 9C = 6A + 8C = 0,\\ x^0: &\quad 2B + 9D = 0. \end{aligned}$$

So $B = 0$ (consistent with $D = 0$), and $6A = -8C = -24$, giving $A = -4$. The required solution is

$$y = 3x - 4x^3.$$

Expressing $\sin 3\theta$

Ask yourself We have two solutions of the same equation: $\sin(3\sin^{-1} x)$ and $3x - 4x^3$. Why must they be the same function?

Both satisfy $(*)$ with $k = 3$, and both satisfy the same conditions at $x = 0$. For the polynomial we built this in. For $y = \sin(3\sin^{-1} x)$ we have $y(0) = \sin 0 = 0$ and, from the first derivative above, $y'(0) = 3\cos 0/1 = 3$.

On $-1 < x < 1$ we may divide $(*)$ by $1 - x^2 \neq 0$. This gives $y'' = \dfrac{xy' - 9y}{1 - x^2}$, a linear second-order equation with continuous coefficients, and such an equation has exactly one solution with a given value of y and y' at $x = 0$. (Think of it as stepping out from $x = 0$: the equation tells you y'' from y and y', so the starting values fix everything.) Hence

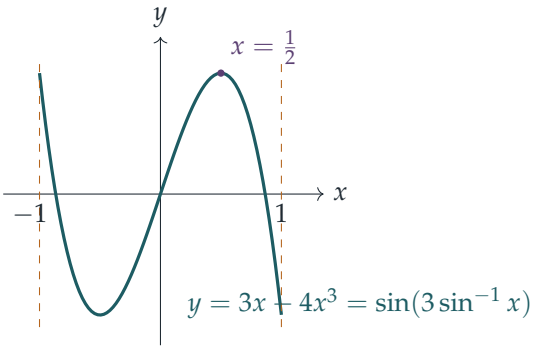
$$\sin(3\sin^{-1} x) = 3x - 4x^3 \qquad \text{for } -1 < x < 1,$$

and by continuity also at $x = \pm 1$.

Now put $x = \sin\theta$ with $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$, so that $\sin^{-1} x = \theta$:

$$\boxed{\sin 3\theta = 3\sin\theta - 4\sin^3\theta.}$$

This holds for every θ, not just $|\theta| \le \frac{\pi}{2}$. Both sides have period 2π. If $\frac{\pi}{2} \le \theta \le \frac{3\pi}{2}$, put $\varphi = \pi - \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then $\sin\varphi = \sin\theta$ and $\sin 3\varphi = \sin(3\pi - 3\theta) = \sin 3\theta$, so the identity for φ gives it for θ.



The graph is the familiar curve $\sin 3\theta$ seen through $x = \sin\theta$: as θ runs from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, 3θ sweeps out one and a half oscillations, hence the two turning points at $x = \pm\frac{1}{2}$ (where $3\theta = \pm\frac{\pi}{2}$).

Checks

- $\theta = \frac{\pi}{2}$: $\sin\frac{3\pi}{2} = -1$ and $3 - 4 = -1$. $\theta = \frac{\pi}{6}$: $\sin\frac{\pi}{2} = 1$ and $\frac{3}{2} - \frac{1}{2} = 1$.
- Numerically, at $\theta = 0.7$: $\sin 2.1 = 0.86321$ and $3(0.64422) - 4(0.64422)^3 = 0.86321$.
- The same identity follows from $\sin(2\theta + \theta)$ and the double-angle formulae, which is a good independent check.

Common slips

- Differentiating $\dfrac{k\cos(k\sin^{-1}x)}{\sqrt{1-x^2}}$ directly with the quotient rule: correct, but messy. Clearing the square root first is much cleaner.
- Forgetting to say *why* the cubic equals $\sin(3\sin^{-1}x)$. Matching the equation alone is not enough: both solutions must also share $y(0)$ and $y'(0)$.

Beyond STEP

Equation $(*)$ is Chebyshev's equation. When $k = n$ is a whole number it has a polynomial solution, and $\cos(n\cos^{-1}x)$ is the Chebyshev polynomial $T_n(x)$, central in numerical approximation. The uniqueness theorem we used (the Picard–Lindelöf theorem) is proved in first-year differential equations courses. Power-series methods for equations like this are treated in Boyce and DiPrima, *Elementary Differential Equations*.

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